How-To Library

Informatica

Configuring Properties for Hive Data Objects in a Mapping

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Abstract

When you create a mapping that includes a Hive data object as the source or target, you can set Hive configuration properties in multiple places. This article describes where you can set Hive configuration properties, the scope of the property based on where it's configured, and the order of precedence that the Data Integration Service follows.

Supported Versions

- Data Engineering Integration 10.2.1 and later

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Overview

When you create a mapping with a Hive data object, you can set Hive configuration properties using the custom properties option for infacmd ms runMapping, in the mapping run-time properties, in the Hadoop connection, in the Hive connection, or in the Data Integration Service custom properties.

Hive configuration properties determine how the Data Integration Service runs jobs that utilize HiveServer2. For example, you can set `tez.task.resource.memory.mb` to optimize the memory allocated to run the job.

If you set the same property in more than one place, the Data Integration Service determines the value based on an override priority. The place that you configure a property determines the scope of its effect. The following table describes the property override priority and the scope:

Where to configure	Priority	Scope
infacmd ms runMapping custom properties	1	Affects only this mapping
Mapping Run-time properties	2	Affects only this mapping
Hadoop Connection Common properties or Hive Pushdown Configuration properties	3	Affects all mappings that use this Hadoop connection
Hive Connection Data Access Connection String or Environment SQL	4	Affects all mappings that use this Hive connection
Data Integration Service custom properties	5	Affects all mappings that this Data Integration Service runs

Infacmd ms runMapping Custom Properties

Set Hive configuration properties using the custom properties option for the infacmd ms runMapping command when the properties apply to only one mapping.

When you run a mapping using the infacmd ms runMapping command, you can configure custom properties for the mapping using the -CustomProperties or -cp option. Enter custom properties as name-value pairs separated by semicolons. To enter a value that contains a space or other non-alphanumeric character, enclose the value in quotation marks.

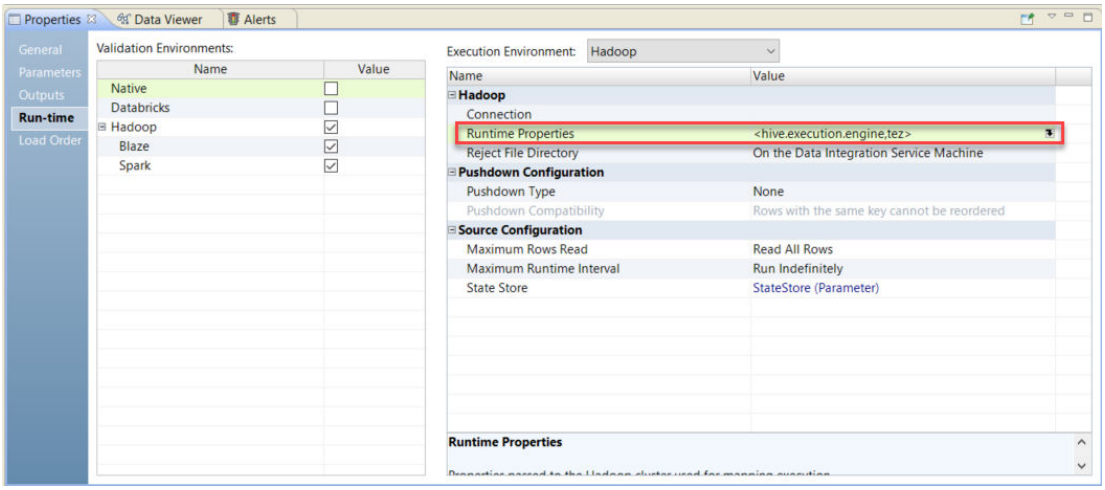
For example, you can include the following phrase when you run infacmd ms runMapping: `-cp hive.execution.engine=tez;tez.task.resource.memory.mb=2048`

Mapping Run-time Properties

Set Hive configuration properties in the mapping run-time properties when the properties apply to only one mapping.

Set the mapping run-time properties in Informatica Developer (the Developer tool). Open a mapping and configure the **Runtime Properties** for the Hadoop execution environment.

The following image shows the property *hive.execution.engine* set to *tez* in the mapping run-time properties:



Hadoop Connection Properties

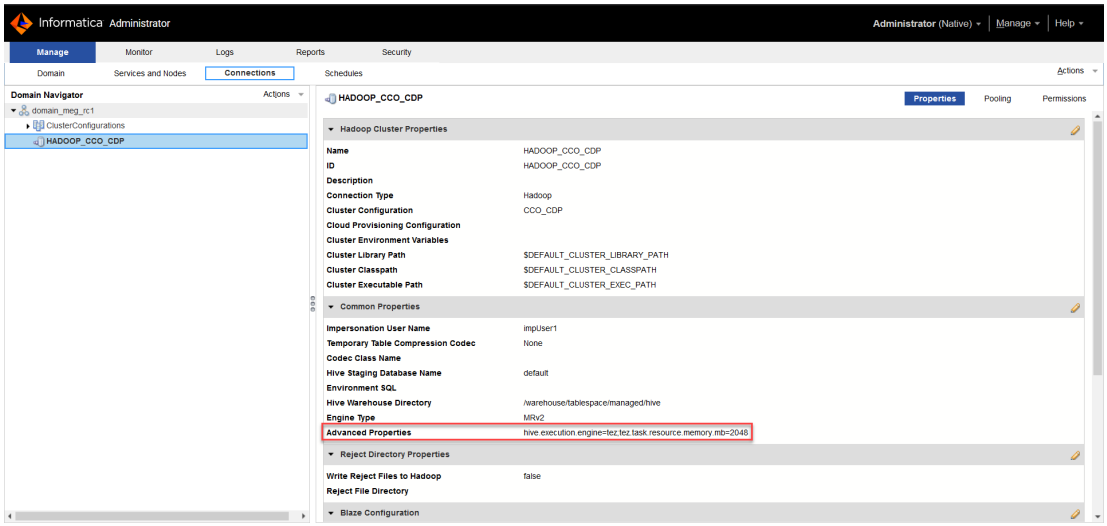
You can set Hive configuration properties in the advanced common properties for the Hadoop connection. Prior to version 10.2.2, you can also set Hive configuration properties in the advanced Hive Pushdown Configuration properties for the Hadoop connection. Configure Hadoop connection properties from Informatica Administrator (the Administrator tool).

If you set the same property with different values in the common properties and in the Hive Pushdown Configuration properties, the Data Integration Service uses the value from the Hive Pushdown Confiugration.

To set a Hive configuration property in the advanced common properties or the advanced Hive Pushdown Configuration properties, use the syntax "<property>:<value>" in a semi-colon separated list. For example, you can append the following string to the list of advanced properties:

```
hive.execution.engine=tez;tez.task.resource.memory.mb=2048
```

The following image shows the advanced properties in the Hadoop connection properties:



Hive Connection Properties

When you set Hive configuration properties in the Hive connection, you can set the properties in either the Data Access Connection String or Environment SQL. Configure Hadoop connection properties from the Administrator tool.

If you set the same property with different values in the Data Access Connection String and Environment SQL, the Data Integration Service uses the value from Environment SQL.

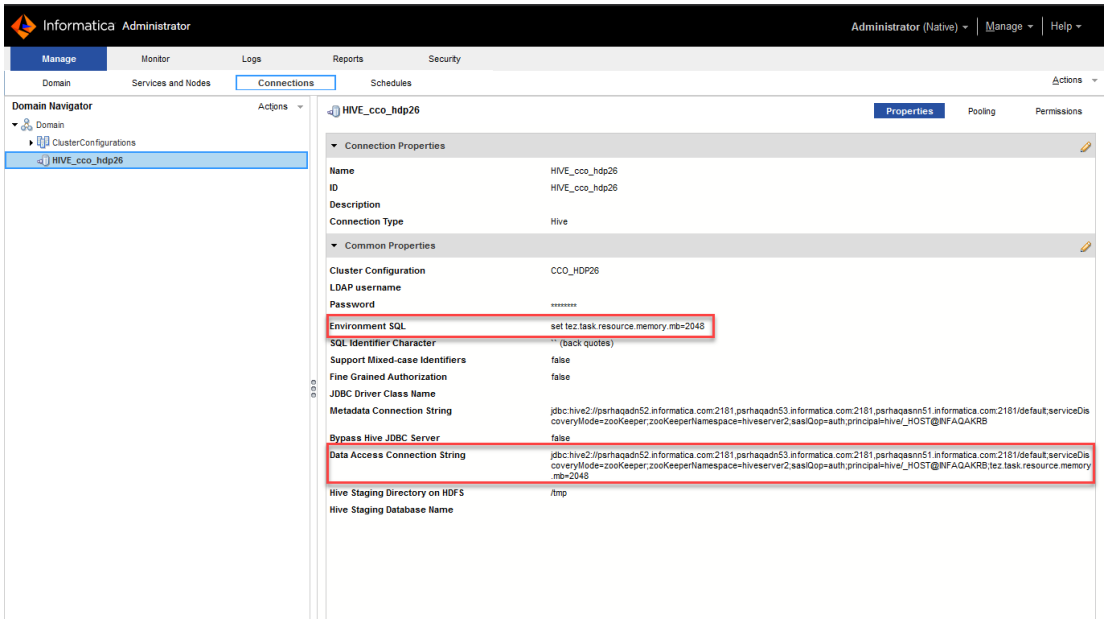
To set a Hive configuration property in the Data Access Connection String, use the syntax "<property>:<value>" in a semi-colon separated list. For example, you can append the following string to the Data Access Connection String:

```
hive.execution.engine=tez;tez.task.resource.memory.mb=2048
```

To set a Hive configuration property in Environment SQL, use the syntax "set <property>=<value>" in a semi-colon separated list. For example, you can append the following string to the Environment SQL:

```
set hive.execution.engine=tez; set tez.task.resource.memory.mb=2048;
```

The following image shows the Environment SQL and Data Access Connection String properties in the Hive connection properties:



Examples of Hive parameters based on different environments

The following table lists some Hive parameters based on different engines/environments and where to use them:

Parameter	Hive Connection property to be used	Engine/Environment
hive.tez.container.size	Data Access Connection String	Native
hive.execution.engine	Data Access Connection String or Environment SQL	Native
tez.task.resource.memory.mb	Data Access Connection String or Environment SQL	Native
hive.stats.column.autogather	Data Access Connection String or Environment SQL	Native
mapreduce.map.memory.mb=2048	Data Access Connection String	Blaze

Data Integration Service Custom Properties

Set Hive configuration properties as Data Integration Service custom properties when the properties apply to any mapping that the Data Integration Service runs.

Configure Data Integration Service custom properties from the Administrator tool. Create a custom property for the Data Integration Service and enter the property name and desired value.

If you configure Hive properties in the mapping run-time properties, Hive connection, or Hadoop connection, they will override the values set in the Data Integration Service.

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